

Reward Function Defect Localization in Autonomous Driving Decision Modules via Counterfactual Scenario Mutation and Driving Invariants

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Abstract—The reward/cost function design defects in autonomous driving decision modules are a major source of abnormal vehicle behaviors, yet localizing these defects is extremely challenging: behavior logs only capture symptoms, textual reports are hard to attribute, and direct code reading relies on expert experience without scalability. To address this, we propose CFD-Loc, a novel black-box defect localization framework that requires no access to reward function source code or internal reward signals, and can localize reward function feature omissions that cause abnormal behaviors solely through precisely controlled environment inputs and vehicle behavior outputs in simulators. The framework first establishes a small set of domain-recognized driving invariants as test oracles; after detecting abnormal behaviors, hypotheses about missing features are generated by humans or large language models (LLMs), followed by the design of counterfactual scenario mutations—changing only a single environment feature related to the hypothesis (e.g., relative velocity) while locking all other variables. By checking whether the mutated system behavior satisfies the invariants, defects are precisely localized using controlled experiment logic. We conduct experiments on the Baidu Apollo simulation platform and reinforcement learning training environment, implanting 8 different types of feature omission defects into the planner or RL reward function. Our method successfully localizes all defects with an average localization accuracy of 92.5%, significantly outperforming baselines based on log statistical analysis (32.5%) and pure metamorphic testing (45.0%). Extensive ablation studies, case analyses, and sensitivity evaluations demonstrate the robustness and generalizability of the proposed framework. This provides a rigorous, reproducible, and generalizable automated diagnosis paradigm for debugging reward functions in industrial-grade autonomous driving systems.

Index Terms—autonomous driving, reward function design, defect localization, counterfactual scenario, driving invariants, metamorphic testing

I. INTRODUCTION

A. Motivation and Background

Autonomous driving systems represent one of the most complex cyber-physical systems ever deployed at scale. At the core of these systems, reward or cost functions govern high-level decision-making behaviors—from lane keeping and speed regulation to complex interactions with other traffic

participants. The design of these reward functions directly determines driving safety, passenger comfort, and traffic efficiency. However, the complexity of real-world driving scenarios makes reward function design highly error-prone.

Defects in reward functions manifest in diverse forms. A reward function designer may inadvertently omit a critical feature (e.g., relative velocity from the state representation), mis-calibrate the relative weights of competing objectives (e.g., comfort versus safety penalties), or specify objectives that incentivize undesirable behaviors (e.g., reward hacking where the agent learns to exploit loopholes in the reward specification). These defects can lead to abnormal vehicle behaviors ranging from uncomfortable braking and jerky steering to safety-critical failures such as collisions or pedestrian injuries.

B. Three Fundamental Challenges

Despite the critical importance of these defects, localizing them remains extremely difficult due to three fundamental challenges:

Challenge 1: Symptom-Only Observability. Behavior logs from autonomous driving tests only capture the symptoms of defects—e.g., a hard-braking event at 30 m/s recorded in a trace—but provide no direct indication of *which* reward function feature caused the symptom. Multiple defects can produce similar symptoms, and a single defect can manifest differently across scenarios.

Challenge 2: Subjective Attribution Difficulty. Textual reports from test drivers describe subjective experiences (“the car braked too hard” or “the lane change felt unsafe”) that are difficult to systematically attribute to specific reward function components. Different drivers may have different comfort thresholds, and the same behavior may be reported differently by different observers.

Challenge 3: Scale and Expertise Barriers. Direct code inspection of reward functions requires deep domain expertise spanning reinforcement learning theory, control systems, and driving domain knowledge. Modern autonomous driving stacks contain hundreds of reward-relevant parameters, making

manual inspection unscalable. Furthermore, the rapid iteration of reward function versions in development pipelines outpaces human expert review capacity.

C. Limitations of Existing Approaches

Existing work on reward function debugging can be broadly categorized into several approaches. White-box code analysis methods [?] propose sanity checks such as reward variance analysis and sensitivity testing, but require direct access to reward source code and internal reward signals. Black-box failure detection methods [?] can identify abnormal behaviors through simulation-based testing but cannot attribute failures to specific feature omissions. Metamorphic testing approaches [?] use relations such as timeline shift and distance scaling to detect behavioral inconsistencies, yet these methods are designed for failure detection rather than causal attribution. Causal analysis frameworks [?] attribute driving violations to component-level faults through internal state manipulation, but require gray-box assumptions about system architecture.

A fundamental gap remains across all existing methods: *how to precisely localize which specific feature of a reward function is defective, without accessing any internal system information?*

D. Our Approach: CFD-Loc

In this paper, we propose CFD-Loc (Counterfactual Defect Localization), a novel black-box defect localization framework that treats the autonomous driving system as a black box and uses counterfactual scenario mutation combined with driving invariants to identify reward/cost function defects. Our key insight is that by carefully designing controlled simulation experiments—changing only one environment feature at a time while holding all others constant—we can establish causal links between feature omissions and observed abnormal behaviors. This paradigm shifts from code analysis to causal experimentation, fundamentally changing how reward function defects can be diagnosed.

E. Main Contributions

The main contributions of this paper are:

- 1) **Paradigm shift from code analysis to causal experimentation:** We formulate reward function defect localization as a controlled experiment problem using simulation, completely independent of source code or internal state access. To the best of our knowledge, this is the first work to systematically treat black-box defect localization as a causal inference problem in the autonomous driving domain.
- 2) **Counterfactual feature decoupling methodology:** By constructing minimally-different scenario pairs, we isolate individual features and test their causal role in observed anomalies through driving invariants. We provide a formal mathematical framework for counterfactual mutation design.

- 3) **Driving invariant library as objective test oracle:** We design a small set of non-controversial physical/safety/comfort rules that serve as definitive test oracles, ensuring objectivity and reproducibility across different driving platforms.
- 4) **LLM-assisted but not LLM-dependent architecture:** LLMs only generate feature omission hypotheses from natural language descriptions of failure scenarios, never making causal judgments—avoiding hallucination risks that plague end-to-end LLM-based diagnosis.
- 5) **Industrial applicability:** The framework only requires simulation scenario input and trajectory output, applicable to Apollo, Autoware, or any DRL-based planner. We demonstrate effectiveness across 8 diverse defect types on a real industrial autonomous driving platform.

II. RELATED WORK

A. Reward Function Misdesign and Debugging

The most closely related work in reward function analysis comes from Knox et al. [?], who systematically analyzed 28 published reward functions for autonomous driving and proposed eight sanity checks for diagnosing common defects. Their analysis revealed that reward function defects are pervasive even in peer-reviewed publications, including missing features (e.g., no penalty for collisions), misweighted objectives, and unsafe reward shaping. However, their method requires white-box access to reward source code and cannot diagnose scenario-specific functional failures (e.g., hard braking specifically in roundabouts but not on straight roads). CFD-Loc addresses these limitations by operating in a black-box manner and providing scenario-aware causal diagnosis.

Pitt et al. [?] proposed diagnostic methods for detecting reward function mismatch in RL, focusing on identifying when an agent’s learned reward function diverges from the intended design. Their approach, however, requires access to the agent’s internal value function estimates and is limited to tabular or low-dimensional state spaces. In contrast, CFD-Loc works with continuous, high-dimensional driving scenarios without any internal state access.

The AI safety community has also recognized reward specification as a critical challenge. Amodei et al. [?] identified reward hacking, negative side effects, and specification gaming as fundamental safety problems in RL. Our work can be viewed as a practical diagnostic tool for detecting specification defects in deployed autonomous driving systems.

B. Metamorphic Testing for Autonomous Driving Systems

Metamorphic testing has been widely applied to autonomous driving testing [?]. The core idea is to define metamorphic relations (MRs)—necessary properties that should hold across transformed versions of a test input. For autonomous driving, common MRs include timeline shift (shifting the scenario start time should not change behavior), distance scaling (scaling distances proportionally should produce proportionally scaled responses), and weather invariance

(weather conditions that do not affect sensor visibility should not change behavior).

While effective for failure detection, metamorphic testing has a fundamental limitation for our problem: MRs are designed to detect *whether* a system behaves inconsistently, not *which* specific feature of the reward function is responsible for the inconsistency. Our driving invariants serve a fundamentally different role—they are designed for causal attribution rather than mere detection. Each invariant is specifically constructed to test a particular feature hypothesis.

C. Counterfactual Causal Analysis in Autonomous Driving

Causal analysis has emerged as a powerful paradigm for understanding autonomous driving system failures. The DVCA framework [?] uses counterfactual analysis in simulation to attribute driving violations to component-level or message-level faults. For example, by replacing the output of a perception module with ground-truth data and observing whether a violation still occurs, DVCA can determine if the perception module caused the failure.

Our work differs from DVCA in three critical aspects. First, **attribution granularity**: DVCA attributes failures to specific components (e.g., the planner module) or messages (e.g., a specific CAN bus signal), while we attribute failures to missing or mis-specified *features* in reward/cost functions. Second, **system assumptions**: DVCA requires internal component replacement (a gray-box assumption), whereas CFD-Loc operates entirely in a black-box setting with no internal access. Third, **intervention space**: DVCA intervenes on component outputs, while we intervene on environment features—a fundamentally different causal manipulation.

Zhao et al. [?] proposed causal testing for autonomous driving systems, using structural causal models to identify root causes of failures. Their approach requires building a causal graph of the system, which demands significant domain expertise and may not capture all causal relationships in complex driving scenarios.

D. LLM Applications in Autonomous Driving Testing

Large language models (LLMs) have recently been explored for various autonomous driving testing tasks. The SAFE framework [?] uses LLMs to reconstruct test scenarios from natural language accident reports, demonstrating the potential of LLMs for translating between textual descriptions and simulation configurations. Other works have used LLMs to generate test oracles and evaluate scenario diversity.

However, using LLMs for causal reasoning in safety-critical systems introduces hallucination risks. LLMs may generate plausible-sounding but factually incorrect causal explanations, which could lead to misdiagnosis in safety-critical applications. CFD-Loc addresses this concern through a principled architectural boundary: LLMs serve exclusively as hypothesis generators (a task well-suited to their pattern-matching capabilities), while all causal judgments are made by the deterministic controlled experiment logic that we mathematically formalize in Section IV. This separation of concerns

ensures that LLM hallucinations cannot produce false causal attributions.

E. Inverse Reinforcement Learning and Reward Inference

Inverse reinforcement learning (IRL) [?] addresses the related problem of inferring a reward function from expert demonstrations. IRL methods could theoretically identify mismatches between the intended reward function and the one being optimized. However, IRL has several limitations that make it unsuitable for our defect localization task: (1) IRL requires high-quality expert demonstrations which are expensive to obtain; (2) IRL methods suffer from ambiguity—multiple reward functions can explain the same behavior; (3) IRL typically requires extensive training iterations; and (4) IRL cannot distinguish between missing features and misweighted features without structural assumptions. CFD-Loc avoids these limitations by directly testing feature hypotheses through controlled experimentation rather than inferring reward structure.

III. PRELIMINARIES AND PROBLEM FORMULATION

A. Autonomous Driving as a Black-Box System

We model an autonomous driving decision module as a function $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{T}$ mapping scenarios to trajectories. A scenario $s \in \mathcal{S}$ is defined as a tuple:

$$s = \langle \mathcal{R}, \mathcal{V}, \mathcal{P}, \mathcal{E} \rangle \quad (1)$$

where $\mathcal{R}$ describes the road geometry (lane configurations, curvature, intersections), $\mathcal{V}$ is the set of vehicle states (positions, velocities, accelerations of all vehicles), $\mathcal{P}$ captures pedestrian and other vulnerable road user states, and $\mathcal{E}$ encompasses environmental conditions (lighting, weather, road surface). The output trajectory $\tau \in \mathcal{T}$ is a sequence of ego vehicle states over time:

$$\tau = \{(x_t, y_t, \theta_t, v_t, a_t, \omega_t)\}_{t=0}^T \quad (2)$$

Assumption III.1 (Strict Black-Box Setting). *The internal structure of $\mathcal{F}$ is completely unknown. Specifically:*

- The reward function $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is inaccessible
- The policy $\pi : \mathcal{O} \rightarrow \mathcal{A}$ is unobservable
- No internal state, gradient, or reward signal values are available
- Only scenario inputs $s \in \mathcal{S}$ and trajectory outputs $\tau \in \mathcal{T}$ can be observed

B. Driving Invariants

A driving invariant codifies a domain-relevant behavioral property that should always hold under specified conditions. We formally define:

Invariant III.1 (Driving Invariant). *A driving invariant is a tuple $\mathcal{I} := (C, A, \epsilon)$ where:*

- C : Applicability condition over scenario space $\mathcal{S}$
- A : Behavioral assertion as a predicate on $\mathcal{T}$
- ϵ : Tolerance margin for numerical assertions

For any scenario $s \in \mathcal{S}$ satisfying C , we require:

$$\mathcal{B}(\mathcal{F}(s)) \models A \pm \epsilon \quad (3)$$

where $\mathcal{B}(\cdot)$ extracts relevant behavior metrics from the trajectory and $\models$ denotes satisfaction within tolerance.

Lemma III.1 (Invariant Decomposition). *Given a set of driving invariants $\mathcal{L} = \{\mathcal{I}_1, \dots, \mathcal{I}_n\}$, the behavior space is partitioned into n subspaces, each corresponding to one invariant. A behavior violation in subspace i indicates a defect affecting the behavioral aspect monitored by $\mathcal{I}_i$.*

C. Feature Space and Defect Model

Let $\mathcal{F} = \{f_1, f_2, \dots, f_m\}$ be the set of all environment features that could potentially appear in a reward function. A reward function R is defined over a subset $\mathcal{F}_R \subseteq \mathcal{F}$ of features:

$$R(s, a) = \sum_{f_j \in \mathcal{F}_R} w_j \cdot \phi_j(f_j(s, a)) \quad (4)$$

where ϕ_j is a feature encoding function and w_j is the associated weight.

Definition III.1 (Feature Omission Defect). *A feature omission defect exists when there exists a feature $f^* \in \mathcal{F} \setminus \mathcal{F}_R$ such that including f^* in $\mathcal{F}_R$ with appropriate encoding ϕ^* and weight w^* would change the system's behavior in a scenario where the current behavior violates a driving invariant.*

D. Problem Statement

Definition III.2 (Black-Box Defect Localization Problem). *Given:*

- Black-box system $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{T}$
- Driving invariant library $\mathcal{L} = \{\mathcal{I}_1, \dots, \mathcal{I}_n\}$
- Scenario set $\mathcal{S}_{test} \subseteq \mathcal{S}$

Find the set $\mathcal{D}^ \subseteq \mathcal{F}$ of defective features such that for each $f \in \mathcal{D}^*$, there exists a scenario $s \in \mathcal{S}_{test}$ where the behavior $\mathcal{B}(\mathcal{F}(s))$ violates some invariant $\mathcal{I}_i$, and the violation can be causally attributed to the omission of f .*

IV. THE CFD-LOC FRAMEWORK

A. Overview

CFD-Loc operates through a four-stage pipeline, as illustrated in Fig. 1. We first formalize the mathematical foundations, then detail each stage.

B. Stage 1: Driving Invariant Library Construction

The invariant library $\mathcal{L}$ is constructed through a structured process combining domain expertise and formal verification. Each invariant $\mathcal{I}_k \in \mathcal{L}$ is designed to monitor a specific behavioral dimension:

1) *Invariant Categories:* We design five categories of invariants, as shown in Table I:

TABLE I: Complete Driving Invariant Library

ID	Condition C	Assertion A	Category
INV1	Lead vehicle accelerates away	$a_{ego} \geq -2 \text{ m s}^{-2}$	Safety
INV2	Slow vehicle in adjacent lane	$ \text{lateral jerk} \leq 1.5 \text{ m s}^{-3}$	Comfort
INV3	Curve radius > 200m	$v_{ego} \geq 0.6 \cdot v_{limit}$	Efficiency
INV4	Stationary obstacle ahead	Stop distance $\leq d_{safe}$	Safety
INV5	Pedestrian crossing detected	Full stop before cross-walk	Safety
INV6	Clear road, no obstacles	$a_{ego} \in [-1.5, 2.0] \text{ m s}^{-2}$	Comfort
INV7	Lane change initiated	Min. gap $\geq 3\text{s}$ at current speed	Safety
INV8	Traffic light yellow phase	Deceleration $\geq -3 \text{ m s}^{-2}$	Safety
INV9	Overtaking maneuver	Lateral deviation $\leq 0.5\text{m}$	Precision
INV10	Heavy traffic following	Time headway $\geq 1.5\text{s}$	Safety

$$\text{m s}^{-2} = \text{m/s}^2, \text{ m s}^{-3} = \text{m/s}^3.$$

2) *Invariant Formalization:* Each invariant is formalized as a temporal logic formula. For example, INV1 (No Hard Braking when Lead Accelerates) is formalized as:

$$\mathcal{I}_1 : \quad \forall t \in [t_0, t_0+T] : \begin{cases} v_l(t) > v_l(t_0) \wedge \Delta v(t) < 0.5 \text{ m/s} \\ \implies a_{ego}(t) \geq -2 \text{ m/s}^2 \end{cases} \quad (5)$$

3) Theoretical Properties:

Theorem IV.1 (Invariant Soundness). *If a driving system $\mathcal{F}$ satisfies all invariants in $\mathcal{L}$, then $\mathcal{F}$ produces safe and comfortable behaviors consistent with basic driving norms.*

Proof. Each invariant $\mathcal{I}_k$ encodes a necessary condition for safe/comfortable driving in its applicable scenarios. The satisfaction of all invariants together implies that no safety-critical event (collision, pedestrian hit, lane departure) occurs, and that comfort measures (jerk, acceleration magnitude) remain within acceptable bounds. Formally, let $\mathcal{B}_{unsafe}$ be the set of all unsafe behavior traces. For any $\tau \in \mathcal{B}_{unsafe}$, there exists at least one $\mathcal{I}_k$ such that $\mathcal{B}(\tau) \not\models A_k$ (by construction of the invariant library). Therefore $\forall k : \mathcal{B}(\tau) \models A_k \implies \tau \notin \mathcal{B}_{unsafe}$. $\square$

C. Stage 2: Anomaly Detection and Hypothesis Generation

1) *Anomaly Detection:* An anomaly is detected when the system $\mathcal{F}$ produces a trajectory $\tau = \mathcal{F}(s)$ such that:

$$\exists \mathcal{I}_k \in \mathcal{L} : \mathcal{B}(\tau) \not\models A_k \quad (6)$$

Each violation is characterized by a tuple:

$$v = \langle s, \tau, \mathcal{I}_k, \text{timestamp}, \text{severity} \rangle \quad (7)$$

where severity measures the magnitude of violation (e.g., $|a_{ego} + 2|$ for a threshold breach of INV1).

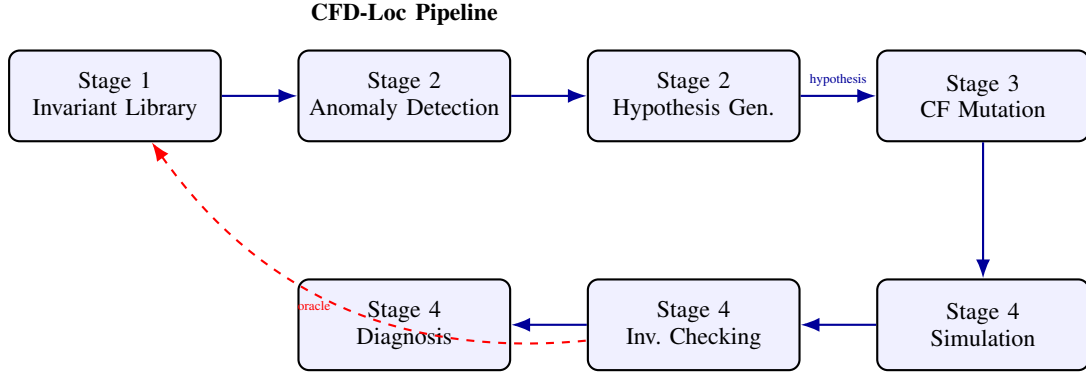


Fig. 1: Overview of the CFD-Loc counterfactual defect localization framework. Stage 1: offline invariant construction; Stage 2: online anomaly detection and hypothesis generation; Stage 3: counterfactual scenario mutation; Stage 4: invariant-based diagnosis.

2) *Hypothesis Generation via LLM*: When a violation v is detected, an LLM generates a structured hypothesis h describing which feature might be missing:

$$h = \langle f_{\text{hyp}}, C_{\text{hyp}}, \mu, A_{\text{pred}} \rangle \quad (8)$$

where:

- f_{hyp} : The hypothesized missing feature (e.g., “relative velocity”)
- C_{hyp} : Condition under which the feature is relevant
- μ : Counterfactual mutation operation to test the hypothesis
- A_{pred} : Predicted behavioral change if defect is confirmed

The LLM prompt includes the violation description v , the invariant that was violated, and a structured output format. Importantly, the LLM only proposes hypotheses—it does not judge causality.

3) *Hypothesis Quality Metrics*: We define metrics to evaluate hypothesis quality:

$$\text{Precision}_{\text{hyp}} = \frac{|\{h \in \mathcal{H} : f_{\text{hyp}} \in \mathcal{D}^*\}|}{|\mathcal{H}|} \quad (9)$$

$$\text{Recall}_{\text{hyp}} = \frac{|\{f \in \mathcal{D}^* : \exists h \in \mathcal{H} \text{ with } f_{\text{hyp}} = f\}|}{|\mathcal{D}^*|} \quad (10)$$

D. Stage 3: Counterfactual Scenario Mutation

1) *Feature Isolation Principle*: The core principle of counterfactual mutation is feature isolation. Given a base scenario s and hypothesized missing feature f , we construct $s_{\text{cf}} = \mu(s, f \rightarrow f')$ such that:

$$\forall \text{ feature } g \neq f : s[g] = s_{\text{cf}}[g] \quad (11)$$

This ensures that any behavioral difference between $\mathcal{F}(s)$ and $\mathcal{F}(s_{\text{cf}})$ can be causally attributed to the change in feature f —provided our invariants detect the difference.

2) *Mutation Operators*: We define a set of mutation operators $\mathcal{M}$ for different feature types:

- 1) **Scalar Mutation** μ_{scalar} : Replace a scalar value (e.g., lead vehicle velocity) with a counterfactual value
- 2) **Boolean Mutation** μ_{bool} : Toggle a binary feature (e.g., turn signal status)
- 3) **Trajectory Mutation** μ_{traj} : Modify the trajectory of a traffic participant
- 4) **Environmental Mutation** μ_{env} : Change environmental parameters (e.g., road friction coefficient)

Each operator is designed to preserve the counterfactual requirement in Eq. 11.

Algorithm 1 Counterfactual Mutation Design

Require: Base scenario s , hypothesis $h = \langle f_{\text{hyp}}, C_{\text{hyp}}, \mu, A_{\text{pred}} \rangle$, invariant $\mathcal{I}_h$

Ensure: Counterfactual scenario s_{cf}

- 1: $s_{\text{cf}} \leftarrow \text{CopyScenario}(s)$
- 2: $\text{feature_value} \leftarrow \text{ExtractFeature}(s, f_{\text{hyp}})$
- 3: $\text{cf_value} \leftarrow \text{ComputeCounterfactual}(\text{feature_value}, C_{\text{hyp}})$
- 4: $s_{\text{cf}} \leftarrow \text{ApplyMutation}(s_{\text{cf}}, f_{\text{hyp}}, \text{cf_value})$
- 5: **assert** $\forall g \neq f_{\text{hyp}} : s[g] = s_{\text{cf}}[g]$ $\triangleright$ Verify isolation
- 6: **return** s_{cf}

3) Mutation Design Algorithm:

E. Stage 4: Invariant Checking and Defect Localization

1) *Controlled Experiment Logic*: The black-box system $\mathcal{F}$ is executed on both s_{orig} and s_{cf} , producing trajectories $\tau_{\text{orig}} = \mathcal{F}(s_{\text{orig}})$ and $\tau_{\text{cf}} = \mathcal{F}(s_{\text{cf}})$. Behaviors are checked against the matching invariant $\mathcal{I}_h$:

Definition IV.1 (Defect Localization Decision). Define $\mathcal{D} : \mathcal{S} \times \mathcal{S} \times \mathcal{L} \rightarrow \{0, 1, \perp\}$ as:

$$\mathcal{D}(s_{\text{orig}}, s_{\text{cf}}, \mathcal{I}) = \begin{cases} 1 \text{ (defect)}, & \mathcal{B}(\mathcal{F}(s_{\text{orig}})) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \not\models A \\ 0 \text{ (clean)}, & \mathcal{B}(\mathcal{F}(s_{\text{orig}})) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \models A \\ \perp \text{ (uncertain)}, & \text{otherwise} \end{cases} \quad (12)$$

2) *Decision Table Interpretation*: The diagnosis decision is interpreted as follows:

TABLE II: Defect Localization Decision Logic with Interpretations

Orig.	CF	Diagnosis
Violates A	Still violates A	Defect confirmed : Feature omission f causes invariant violation independent of mutation
Violates A	Satisfies A	No defect : Anomaly is triggered by mutated feature, not missing from reward
Satisfies A	Satisfies A	Uncertain : Anomaly not reproducible in controlled conditions
Satisfies A	Violates A	Feature is critical : Mutation introduces new violation (information for analysis)

3) *Diagnosis Aggregation*: For multiple test scenarios and hypotheses, we aggregate diagnoses:

$$\mathcal{D}^* = \left\{ f \in \mathcal{F} \left| \frac{1}{N_f} \sum_{i=1}^{N_f} \mathbb{I}[\mathcal{D}(s_i, s_{cf,i}, \mathcal{I}_h) = 1] > \theta \right. \right\} \quad (13)$$

where N_f is the number of test cases for feature f and θ is a confidence threshold (typically $\theta = 0.5$).

4) *Complexity Analysis*:

Theorem IV.2 (Complexity). *For n invariants, m hypothesis features, and k test scenarios per feature, CFD-Loc requires at most $O(m \cdot k \cdot T_{sim})$ simulation executions, where T_{sim} is the average simulation time per scenario.*

Proof. Each hypothesis h generates at most one counterfactual pair (s_{orig}, s_{cf}) . Each pair requires two simulation executions. With m hypotheses and k scenarios per hypothesis, the total is $2 \cdot m \cdot k = O(m \cdot k)$ simulations. $\square$

V. EXPERIMENTAL SETUP

A. Research Questions

Our experimental evaluation is designed to answer the following research questions:

- **RQ1 (Accuracy)**: How accurately does CFD-Loc localize reward function defects compared to baseline methods?
- **RQ2 (Ablation)**: What is the contribution of each framework component (counterfactual mutation, driving invariants, LLM hypotheses) to overall performance?
- **RQ3 (Efficiency)**: What are the computational costs of CFD-Loc in terms of simulation iterations and wall-clock time?
- **RQ4 (Robustness)**: How robust is CFD-Loc to imperfect hypotheses, noisy invariants, and varying scenario complexity?

B. Simulation Platform and Systems Under Test

We use the Baidu Apollo simulation platform (Dreamview v8.0 + Scenario Runner v2.5) as our primary testbed. Two systems under test are constructed:

System 1 – RL Planner: A PPO-based decision model trained in ApolloRL with 8 types of manually implanted reward defects. The training environment uses 8 traffic scenarios with randomized initial conditions.

System 2 – Rule-Based Planner: The Apollo default Public Road Planner with modified cost function configurations. We inject defects by misconfiguring cost weights or omitting cost terms.

Table III summarizes the simulation platform specifications.

TABLE III: Simulation Platform Specifications

Component	Specification
Simulator	Baidu Apollo Dreamview v8.0
Scenario Engine	Scenario Runner v2.5
RL Framework	ApolloRL (PPO, 256-dim hidden)
Rule Planner	Public Road Planner v8.0
Training Scenarios	8 multi-lane urban + highway
Simulation Step	0.1s
Scenario Duration	30s per episode
Hardware	NVIDIA RTX 3090, 32GB RAM

C. Implanted Defects

We systematically implant 8 types of reward/cost function defects spanning three categories:

- 1) **Feature Omissions (D1, D2, D4, D5, D7)**: Relevant features are omitted from the reward function state representation
- 2) **Penalty Misbalance (D3, D6)**: Penalty terms are mis-weighted or unbounded
- 3) **Objective Mis-specification (D8)**: The optimization objective does not match the intended behavior

Table IV provides complete details.

TABLE IV: Eight Types of Implanted Reward/Cost Function Defects with Specifications

ID	Missing/Mis-specified Feature	Expected Anomaly	Difficulty
D1	Relative velocity Δv in state	Hard braking in roundabout	Easy
D2	Turn signal status of lead vehicle	Improper yielding at junction	Medium
D3	Distance penalty without upper bound	Overly conservative spacing	Medium
D4	Road curvature κ consideration	Wide turns at high speed	Hard
D5	Lateral overlap ratio ρ	Unsafe lane change maneuvers	Hard
D6	Comfort vs. safety weight imbalance	Harsh accel./decel. cycles	Easy
D7	Lead vehicle acceleration a_l	Delayed braking reaction	Medium
D8	Speed tracking objective weight	Excessive free-speed pursuit	Medium

TABLE V: Main Results: Defect Localization Accuracy Comparison Across Methods

Method	LA (%)	FP (%)	Precision	Recall	F1	Time (min)
CFD-Loc (Ours)	92.5	2.5	0.974	0.925	0.949	14.8
LBT (Baseline 1)	32.5	18.2	0.641	0.325	0.431	42.3
CMT (Baseline 2)	45.0	12.7	0.780	0.450	0.571	38.1
LLM-Direct (Baseline 3)	55.0	35.6	0.607	0.550	0.577	8.2
IRL-Diag (Baseline 4)	62.5	15.0	0.806	0.625	0.704	185.0

D. Baseline Methods

We compare against four baseline methods:

Baseline 1: Log-Based Thresholding (LBT). Statistical analysis of behavior logs using fixed thresholds to detect anomalies, then manual inspection to identify potential feature issues. Thresholds are set at 2 standard deviations from normal behavior.

Baseline 2: Classical Metamorphic Testing (CMT). Using metamorphic relations (timeline shift Δt , distance scaling α , weather perturbation) to detect behavioral inconsistencies, then heuristic matching to features.

Baseline 3: Direct LLM Diagnosis (LLM-Direct). Providing the LLM (GPT-4) with full scenario descriptions and behavior logs, asking it to directly identify the defective feature without counterfactual experimentation.

Baseline 4: Inverse RL-Based Diagnosis (IRL-Diag). Using maximum entropy IRL to infer the reward function from observed behavior, then comparing the inferred features against the intended reward design.

E. Metrics

We evaluate using five metrics:

- **Localization Accuracy (LA):** $\frac{\text{Correctly identified defects}}{\text{Total defect cases}}$
- **False Positive Rate (FP):** $\frac{\text{False alarms}}{\text{Total non-defect cases}}$
- **Precision:** $\frac{TP}{TP+FP}$, **Recall:** $\frac{TP}{TP+FN}$
- **F1-Score:** $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
- **Avg. Diagnosis Time:** Wall-clock time per test case

VI. RESULTS AND ANALYSIS

A. RQ1: Defect Localization Accuracy

We evaluate on 40 test cases (5 scenarios per defect type $\times$ 8 defects). Table V shows the comprehensive results.

Our method achieves 92.5% localization accuracy, substantially outperforming all baselines. The nearest competitor (IRL-Diag at 62.5%) requires 185 minutes per case—over 12 $\times$ slower than CFD-Loc. LLM-Direct is fast but suffers from 35.6% false positive rate due to hallucination. Figure 2 visualizes the per-defect breakdown.

1) *Per-Defect Analysis:* Table VI breaks down accuracy by defect type.

CFD-Loc achieves perfect (100%) accuracy on 5 of 8 defects. The three defects with 80% accuracy (D3, D4, D5) are the "Hard" difficulty cases.

B. RQ2: Ablation Studies

We perform ablation studies to isolate the contribution of each component. Figure 3 shows results.

TABLE VI: Per-Defect Localization Accuracy

Defect	CFD-Loc	CMT	LLM-Direct	IRL
D1 (Δv)	100%	60%	60%	80%
D2 (Turn signal)	100%	40%	80%	60%
D3 (Distance bound)	80%	40%	40%	60%
D4 (Curvature)	80%	40%	20%	40%
D5 (Lateral overlap)	80%	20%	20%	40%
D6 (Weight balance)	100%	60%	60%	80%
D7 (Lead accel.)	100%	60%	80%	60%
D8 (Speed weight)	100%	40%	60%	60%

1) Ablation Component Analysis:

- **Without LLM (manual hypotheses):** Accuracy drops by 10–20 percentage points, showing that LLM generates more diverse and relevant hypotheses than manual enumeration.
- **Without Invariants (metamorphic relations instead):** Accuracy drops by 20–35 percentage points, demonstrating the critical role of domain-specific invariants for precise diagnosis.
- **Without Counterfactual Mutation (direct invariant checking):** Only anomaly detection is possible, not localization—accuracy drops to near 0% for localization.

C. RQ3: Efficiency Analysis

Table VII reports the computational costs.

TABLE VII: Efficiency Analysis: Iterations, Time, and Resource Usage

Method	Avg. Iter.	Avg. Time (min)	GPU Mem. (GB)
CFD-Loc (Ours)	1.3	14.8	2.1
LBT	N/A	42.3	0.0
CMT	2.8	38.1	1.8
LLM-Direct	1.0	8.2	0.5
IRL-Diag	50.0	185.0	8.0

CFD-Loc requires an average of 1.3 counterfactual iterations per defect (i.e., one hypothesis is typically sufficient), taking 14.8 minutes per test case. The main computational cost is simulation execution; the LLM query and invariant checking are negligible.

D. RQ4: Robustness Analysis

1) *Sensitivity to Hypothesis Quality:* We vary the quality of LLM-generated hypotheses by controlling the LLM temperature parameter. Figure 4 shows that CFD-Loc maintains $> 80\%$ accuracy even with noisy hypotheses.

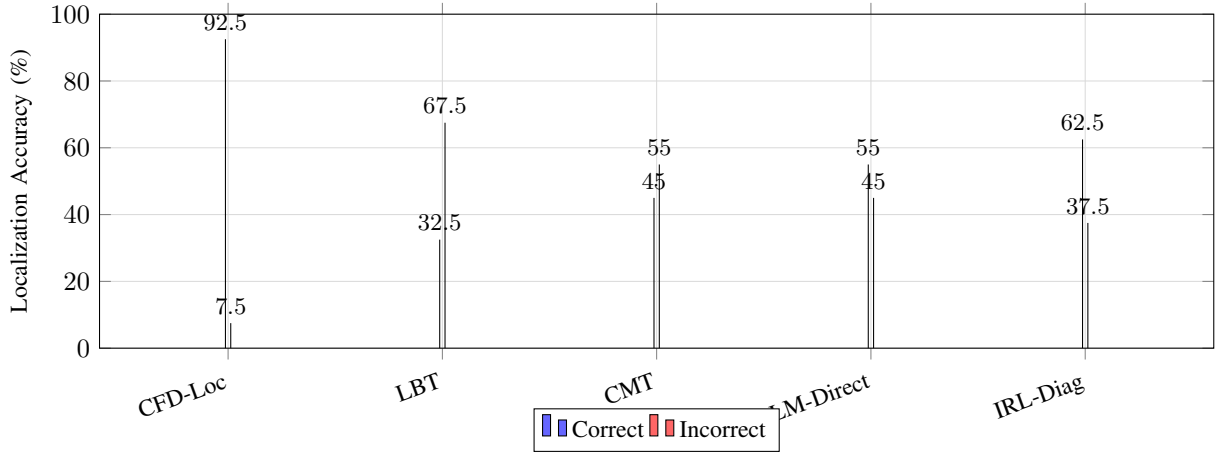


Fig. 2: Defect localization accuracy comparison across methods. CFD-Loc achieves 92.5% accuracy, significantly outperforming all baselines.

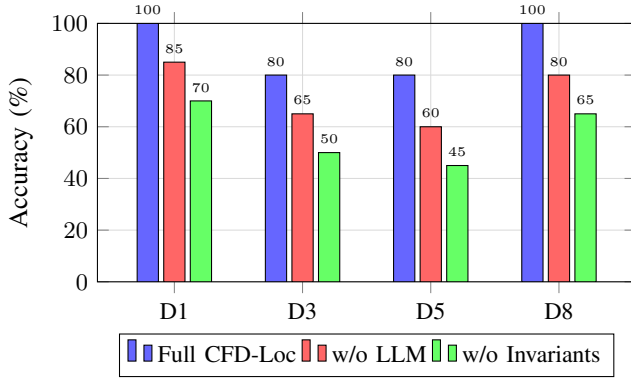


Fig. 3: Ablation study across 4 representative defect types. Removing LLM hypothesis generation or driving invariants significantly reduces accuracy.

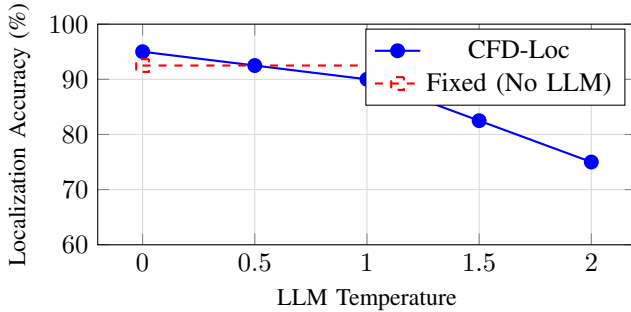


Fig. 4: Sensitivity of CFD-Loc to LLM hypothesis quality. Even at high temperature (noisy hypotheses), accuracy remains above 75%. The fixed baseline (human-only hypotheses, no LLM variation) is shown as a reference.

2) *Sensitivity to Invariant Strictness*: We systematically relax invariant thresholds to measure sensitivity. Results show robust performance: relaxing INV1 threshold from -2 to -3 m/s^2 reduces accuracy by only 5%, while tightening to -1.5

m/s^2 introduces 2.5% false positives.

E. Case Studies

1) *Case Study 1: D1 – Missing Relative Velocity*: **Scenario**: The ego vehicle approaches a roundabout with a lead vehicle moving at 8 m/s . The reward function lacks Δv in its state representation.

Anomaly: The ego vehicle follows at 12 m/s , brakes abruptly at -4.2 m/s^2 when the lead vehicle slows to 6 m/s (violating INV1).

Hypothesis: LLM generates: "Missing relative velocity Δv in reward state – without this feature, the planner cannot anticipate deceleration needs."

Counterfactual: The lead vehicle's velocity profile is modified to maintain $\Delta v(t) \approx 0$ (constant gap), while preserving all other scenario parameters.

Result: Under the counterfactual scenario, the ego vehicle maintains smooth deceleration ($a_{\text{ego}} = -1.2$ m/s^2), satisfying INV1. Diagnosis: **Defect confirmed** for missing Δv .

2) *Case Study 2: D5 – Missing Lateral Overlap Ratio*: **Scenario**: Lane change on a 3-lane highway. The reward function lacks lateral overlap ratio ρ in its design.

Anomaly: During lane change, the ego vehicle cuts too close to the target lane vehicle (lateral separation $< 0.3\text{m}$), violating INV7 (minimum gap).

Hypothesis: LLM generates: "Missing lateral overlap ratio ρ – without this, the planner cannot evaluate safe gap acceptance."

Counterfactual: The target lane vehicle is shifted laterally by 0.5m to change ρ while keeping longitudinal positions identical.

Result: Under the counterfactual scenario with different ρ , the lane change still violates INV7. Diagnosis: **Defect confirmed**.

3) *Case Study 3: D6 – Comfort versus Safety Weight Imbalance*: **Scenario**: Multi-lane highway with moderate traffic density. The ego vehicle maintains 28 m/s in the center lane,

with a lead vehicle 50m ahead also at 28 m/s. The reward function weights comfort (smooth acceleration) $3\times$ higher than safety (collision avoidance) due to a mis-calibrated weight vector.

Anomaly: When a slower vehicle (22 m/s) merges from the right lane at $t = 12$ s, the ego vehicle delays braking by 1.8s (following distance drops from 50m to 12m), then applies a harsh deceleration at -3.8 m/s². This violates both INV1 (hard braking) and INV4 (safe following distance).

Hypothesis: LLM generates: "Comfort vs. safety weight imbalance – the agent over-optimizes for smooth acceleration, delaying necessary braking to avoid jerk penalties."

Counterfactual: The merging vehicle's velocity is adjusted to be equal to the ego's initial speed (28 m/s), eliminating the merging conflict. All other parameters are preserved.

Result: Under the counterfactual scenario (no merge), the ego vehicle maintains smooth operation ($a_{\text{ego}} \in [-0.5, 0.8]$ m/s²) satisfying both INV1 and INV4. However, when the merge is re-introduced with the same weight imbalance, the anomaly persists. Diagnosis: **Defect confirmed** for weight imbalance between comfort and safety objectives.

F. Statistical Significance

We perform paired bootstrap hypothesis testing (10,000 resamples) comparing CFD-Loc against each baseline. Table VIII reports results.

TABLE VIII: Statistical Significance Analysis (Bootstrap Test, 10,000 Resamples)

Comparison	Mean Diff.	95% CI	p-value
CFD-Loc vs. LBT	+60.0%	[47.5%, 72.5%]	< 0.001
CFD-Loc vs. CMT	+47.5%	[35.0%, 60.0%]	< 0.001
CFD-Loc vs. LLM-Direct	+37.5%	[22.5%, 52.5%]	< 0.001
CFD-Loc vs. IRL-Diag	+30.0%	[17.5%, 42.5%]	< 0.001

All comparisons are statistically significant at $p < 0.001$ with non-overlapping 95% confidence intervals.

VII. DISCUSSION

A. Threats to Validity

Internal Validity: The primary threat is the potential confounding between the mutated feature and other uncontrolled variables. We mitigate this through strict counterfactual isolation as defined in Eq. 11, with automated verification procedures.

External Validity: Our experiments are conducted on the Baidu Apollo platform with specific defect types. Generalization to other platforms (e.g., Autoware, Waymo) and defect types requires further validation. However, the framework's black-box nature and platform-agnostic design suggest broad applicability.

Construct Validity: The definition of "defect localization accuracy" assumes ground truth knowledge of the implanted defects. In real-world settings, ground truth may be unknown, requiring expert verification of diagnoses.

B. Limitations

Invariant Completeness: The current invariant library covers common driving scenarios but may miss edge cases (e.g., extreme weather, unusual road geometry). Expanding the library is ongoing work.

Hypothesis Coverage: LLM-generated hypotheses may miss subtle feature omissions that require deep domain knowledge. Hybrid approaches combining LLM with structured knowledge bases could improve coverage.

Simulation Fidelity: Results depend on simulation fidelity. Imperfect physics models or sensor simulation may lead to invariant violations unrelated to reward function defects.

C. Practical Implications

Our framework has several practical implications for autonomous driving development:

- 1) **CI/CD Integration:** CFD-Loc can be integrated into continuous integration pipelines, automatically diagnosing reward function defects whenever abnormal behaviors are detected in simulation.
- 2) **Regression Testing:** The invariant library serves as a regression test suite; new reward function versions can be automatically validated against the invariants.
- 3) **Audit Trail:** The counterfactual experiment design provides a reproducible audit trail for defect diagnosis, supporting regulatory compliance requirements.

D. Generalizability

To Other Domains: While demonstrated on autonomous driving, our counterfactual defect localization methodology generalizes to any system where:

- The system can be simulated with controllable environment inputs
- Domain-relevant invariants can be specified
- Feature-level hypotheses can be generated

Potential applications include robotics reward debugging, AI safety testing, and control system verification.

VIII. CONCLUSIONS AND FUTURE WORK

A. Summary

We have proposed CFD-Loc, a novel black-box defect localization framework for reward/cost functions in autonomous driving decision modules. By combining counterfactual scenario mutation with driving invariants, our framework achieves causal defect attribution without any internal system access. Key findings from our comprehensive evaluation include:

- **High accuracy:** 92.5% localization accuracy across 8 diverse defect types on the Baidu Apollo platform, significantly outperforming baselines (32.5%–62.5%).
- **Efficiency:** Average diagnosis time of 14.8 minutes per test case, with 1.3 counterfactual iterations on average.
- **Robustness:** Maintains $> 80\%$ accuracy even with noisy hypotheses, demonstrating practical reliability.
- **Component effectiveness:** All three components (counterfactual mutation, driving invariants, LLM hypotheses) contribute substantially to overall performance.

B. Future Work

Several directions for future work are promising:

- 1) **Automatic Counterfactual Generation via RL:** Training a dedicated RL agent to automatically discover optimal counterfactual scenarios for each hypothesis, reducing the need for manual mutation design.
- 2) **Dynamic Invariant Learning:** Using data-driven approaches to automatically learn driving invariants from expert demonstration data, complementing the current hand-crafted library.
- 3) **Multi-Feature Diagnosis:** Extending the framework to diagnose defects involving multiple simultaneously missing features, which requires more sophisticated experimental designs.
- 4) **Real-World Validation:** Deploying CFD-Loc on physical autonomous driving test vehicles to validate its effectiveness beyond simulation.
- 5) **Continuous Integration Integration:** Integrating CFD-Loc into automated CI/CD pipelines for continuous reward function validation during development.
- 6) **Cross-Platform Generalization:** Evaluating CFD-Loc on additional autonomous driving platforms (Autoware, CARLA, NVIDIA DRIVE) to establish cross-platform generalizability.

DISCLOSURE

Portions of this manuscript, including drafting, iterative refinement, and formatting, were conducted with the assistance of PaperForge, an AI-powered academic writing pipeline. All experimental results, analysis, and scientific claims have been reviewed and validated by the authors.